

## Geometric series



- 1 Find the sum of the first 12 terms of the series:  $5 + 10 + 20 \dots$
- 2 Find the sum of the first 5 terms of the geometric sequence defined by the following. Round your answers to two decimal places.
  - a  $a = 2.187$  and  $r = 1.134$
  - b  $a = -4.186$  and  $r = -2.848$
- 3 Find the sum of the first 11 terms of the series:  $1 - 2 + 4 \dots$
- 4 Find the sum of the first 7 terms of the geometric series:  $64 + 16 + 4 \dots$
- 5 Find the sum of the first 5 terms of the series:  $24 - 12 + 6 \dots$   
Round your answer to the nearest whole number.
- 6 Find the sum of the first 7 terms of the series:  $8 + 1 + \frac{1}{8} \dots$   
Round your answer to three decimal places.
- 7 Consider the series:  $5 + \frac{5}{2} + \frac{5}{4} \dots$ 
  - a Find the common ratio,  $r$ .
  - b Find the sum of the first 10 terms, rounding your answer to one decimal place.
- 8 The first 3 terms of a sequence are  $x, \frac{2}{5}x^2, \frac{4}{25}x^3$ .
  - a Write a simplified expression for the sum of the first  $n$  terms of this sequence.
  - b Evaluate the sum when  $x = 5$  and  $n = 9$ .
- 9 Consider the series:  $\frac{1}{4} - \frac{1}{5} + \frac{1}{16} - \frac{1}{25} + \frac{1}{64} - \frac{1}{125} + \dots$   
Form an expression for the sum of the first  $2n$  terms of the series.
- 10 For each of the following series:
  - i Find  $n$ , the number of terms in the series.
  - ii Find the sum of the series.



- 11 Find  $n$ , the number of terms, in each of the following series:
- a The sum of  $n$  terms in the geometric series  $2 + 10 + 50 + \dots$  is 195 312.
  - b The sum of  $n$  terms in the geometric series  $5, -20, 80, \dots$  is 262 145.
  - c The sum of  $n$  terms in the geometric series  $16 + 4 + 1 + \dots$  is  $\frac{1365}{64}$ .
  - d The sum of  $n$  terms in the geometric series  $40, -8, \frac{8}{5}, \dots$  is  $\frac{20\,832}{625}$ .
- 12 The sum of the first 6 terms of a geometric series is 28 times the sum of its first 3 terms. Find  $r$ , the common ratio.
- 13 The sum of the first 9 terms of a geometric series is 436 905, and the common ratio is  $\frac{1}{4}$ . Find  $a$ , the first term in the series.
- 14 The first term of a geometric sequence is 9 and the fourth term is 243. What is the sum of the first 13 terms of the corresponding series?
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## Infinite series

- 15 What condition must be satisfied by an infinite geometric series in order for its sum to exist?
- 16 For each of the following infinite geometric sequences:
- i Find the common ratio.
  - ii Determine whether the sum is divergent or convergent.
- |                          |                         |
|--------------------------|-------------------------|
| a 11, 22, 44, 88, ...    | b 3, -12, 48, -192, ... |
| c -40, -20, -10, -5, ... | d 256, 64, 16, 4, ...   |
- 17 For each of the following infinite geometric sequences:
- |                                                      |                                               |
|------------------------------------------------------|-----------------------------------------------|
| i Find the common ratio, $r$ .                       | ii Find the limiting sum of the series.       |
| a $2, \frac{1}{2}, \frac{1}{8}, \frac{1}{32}, \dots$ | b 125, 25, 5, 1, ...                          |
| c 16, -8, 4, -2, ...                                 | d $28, -7, \frac{7}{4}, -\frac{7}{16}, \dots$ |



18 For a particular geometric sequence,  $a = 2$  and  $a_{\infty} = 4$ .

a Find  $r$ , the common ratio.

b Find the first 3 terms in the sequence.

19 Consider the infinite sequence:  $1, x - 7, (x - 7)^2, (x - 7)^3, \dots$

Determine the range of values of  $x$  if the infinite series has a limiting sum.

20 The limiting sum of the infinite sequence  $1, 4x, 16x^2, \dots$  is 5. Find the value of  $x$ .

21 Consider a geometric series in which all terms are positive that has first term  $a$  and common ratio  $r$ . The sum of the infinite series is 768. The sum of the first 4 terms is 765.

Find the first three terms of the sequence.

22 Consider the infinite geometric series:  $6 + 2 + \frac{2}{3} + \frac{2}{9} + \dots$

a Find the number of terms,  $n$ , that would be required to give a sum of  $\frac{177\,146}{19\,683}$ .

b Find the infinite sum of the series.

23 Consider the infinite geometric series:  $5 + \sqrt{5} + 1 + \dots$

a Find the common ratio,  $r$ , expressing your answer with a rational denominator.

b Find the limiting sum, expressing your answer with a rational denominator.

24 Find the limiting sum of the infinite series  $\frac{1}{5} + \frac{3}{5^2} + \frac{1}{5^3} + \frac{3}{5^4} + \frac{1}{5^5} + \dots$

## Sigma notation

25 Evaluate:

a

$$\sum_{i=1}^4 4^i$$

b

$$\sum_{j=1}^5 243 \left(\frac{5}{3}\right)^j$$

c

$$\sum_{j=1}^5 32 \left(\frac{1}{2}\right)^j$$

d

$$\sum_{i=3}^8 (-3)^i$$

e

$$\sum_{i=2}^7 (-2^i)$$

f

$$\sum_{i=1}^5 (2(3)^{i-1})$$

g

$$\sum_{j=4}^8 \left(\frac{1}{4}(3)^{j-1}\right)$$

h

$$\sum_{j=1}^4 \left(-3 \left(\frac{1}{3}\right)^j\right)$$

26 Write the series  $1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \dots + \frac{1}{1024}$  in sigma notation.



27 Find the value of the following:

a

$$\sum_{i=1}^{\infty} 3 \left( \frac{1}{4} \right)^{i-1}$$

b

$$\sum_{i=1}^{\infty} 5 \left( -\frac{1}{4} \right)^{i-1}$$

c

$$\sum_{i=1}^{\infty} \frac{1}{8} \left( -\frac{1}{2} \right)^{i-1}$$

d

$$\sum_{i=1}^{\infty} -\frac{1}{2} \left( \frac{5}{7} \right)^{i-1}$$

e

$$\sum_{i=1}^{\infty} (0.9)^i$$

f

$$\sum_{k=1}^{\infty} 9^{-k}$$

## Repeating decimals

28 The repeating decimal  $0.8888\dots$  can be expressed as a fraction when viewed as an infinite geometric series.

- a Using fractions, write the first five terms of the geometric sequence representing  $0.8888\dots$
- b State the values of the first term  $a$  and the common ratio  $r$  of this sequence.
- c Calculate the infinite sum of the sequence as a fraction.

29 The repeating decimal  $0.444444\dots$  can be expressed as a fraction when viewed as an infinite geometric series.

- a Express the first two decimal places,  $0.44$ , as an unsimplified fraction.
- b Express the second two decimal places,  $0.0044$ , as an unsimplified fraction.
- c Express the third two decimal places,  $0.000044$ , as an unsimplified fraction.
- d Now, state the values of the first term  $a$  and the common ratio  $r$  of the sequence formed from these first three terms.
- e Calculate the infinite sum of the sequence as a fraction.

30 For each of the following repeating decimals:

- i Rewrite the decimal as a geometric series.
- ii Express the decimal as a fraction.

a  $0.6666\dots$

b  $0.06666\dots$

31 Consider the sum  $0.25 + 0.0025 + 0.000025$



a Express the sum as a decimal.

b Now, express the sum as a fraction.

32 Consider the repeating decimal  $0.2222\ldots$  in the form

$$\frac{2}{10} + \frac{2}{100} + \frac{2}{1000} + \frac{2}{10\,000} + \ldots$$

Rewrite the repeating decimal as a fraction.

33 Consider the repeating decimal  $0.575\,75\ldots$  in the form

$$0.57 + 0.0057 + \ldots$$

Rewrite the repeating decimal as a fraction.

34 Find the value of  $0.994 + 0.000\,994 + 0.000\,000\,994\ldots$

## Applications

35 At the start of 2014, Pauline deposits \$5000 into an investment account. At the end of each quarter, she makes an extra deposit of \$700.

The table below shows the first few quarters of 2014. All values in the table are in dollars:

| Quarter   | Opening Balance | Interest | Deposit | Closing Balance |
|-----------|-----------------|----------|---------|-----------------|
| Jan - Mar | 5000            | 200      | 700     | 5900            |
| Apr - Jun | 5900            | 236.00   | 700     | 6836.00         |
| Jul - Sep | 6836.00         | 273.44   | 700     | 7809.44         |

a Find the quarterly interest rate.

b Write an expression for the amount in the account at the end of the first quarter.

c Write an expression for the amount in the account at the end of the second quarter.

d Write a similar expression for the amount in the account at the end of the third quarter.

e Write an expression for the amount in the investment account after  $n$  quarters.

f Now, determine the total amount in Pauline's account at the beginning of 2016 to the nearest dollar.





- 39** Rochelle invests \$190 000 at a rate of 7% per annum compounded annually, and wants to work out how much she can withdraw each year to ensure the investment lasts 20 years.
- a** Write an expression for  $A_1$ , the balance of the account after 1 year. Use  $x$  to represent the amount to be withdrawn each year.
  - b** Write an expression for:
    - i**  $A_2$
    - ii**  $A_3$
  - c** Write an expression for  $A_n$ .
  - d** Now, determine Rochelle's annual withdrawal amount.
- 40** Mario invests \$110 000 at a rate of 16% per annum compounded quarterly and wants the investment to last 25 years.
- a** The amount in the account after  $n$  quarters can be expressed as the  $n$ th term of a geometric sequence minus the sum of a different geometric sequence.  
Write an expression for the amount in the investment account after  $n$  quarters. Use  $x$  to represent the amount to be withdrawn each quarter.
  - b** Now, determine Mario's quarterly withdrawal amount.
- 41** Emma invests \$190 000 at a rate of 3% per annum compounded monthly, and wants to work out how much she can withdraw each month to ensure the investment lasts 30 years.
- a** The amount in the account after  $n$  months can be expressed as the  $n$ th term of a geometric sequence minus the sum of a different geometric sequence.  
Write an expression for the amount in the investment account after  $n$  months. Use  $x$  to represent the amount to be withdrawn each month.
  - b** Now, determine Emma's monthly withdrawal amount.